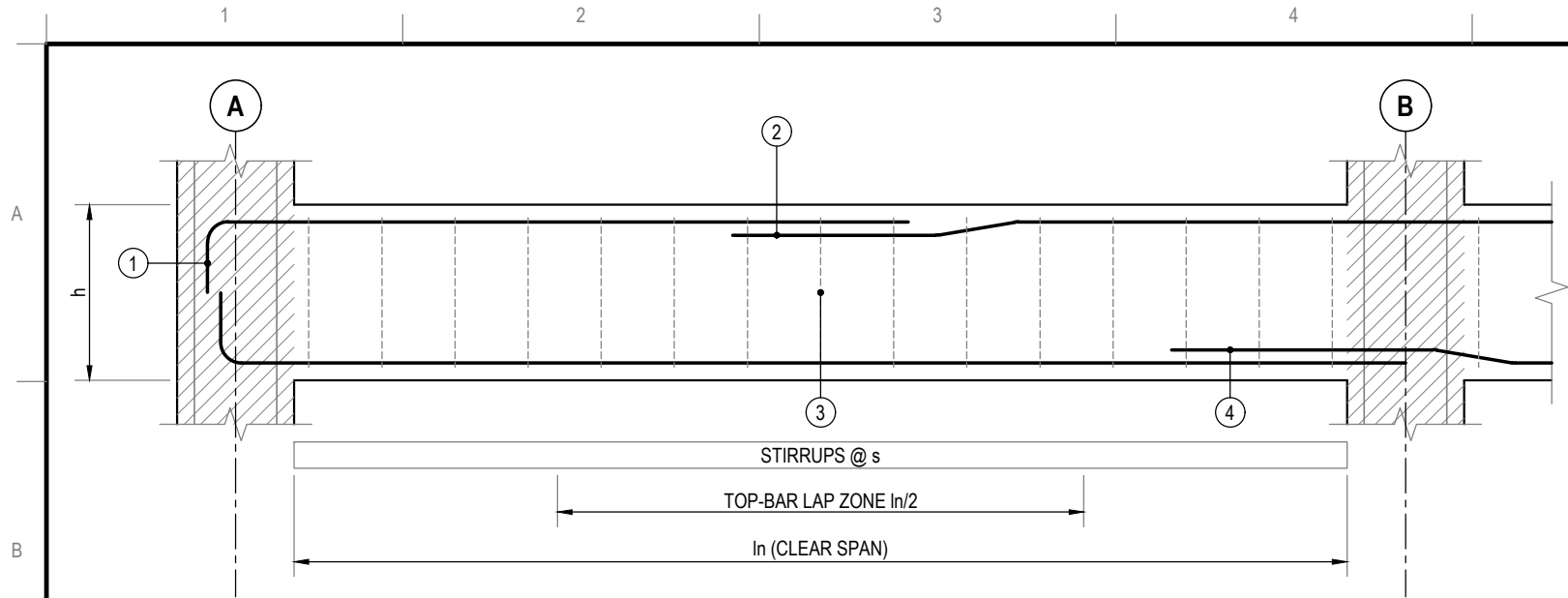
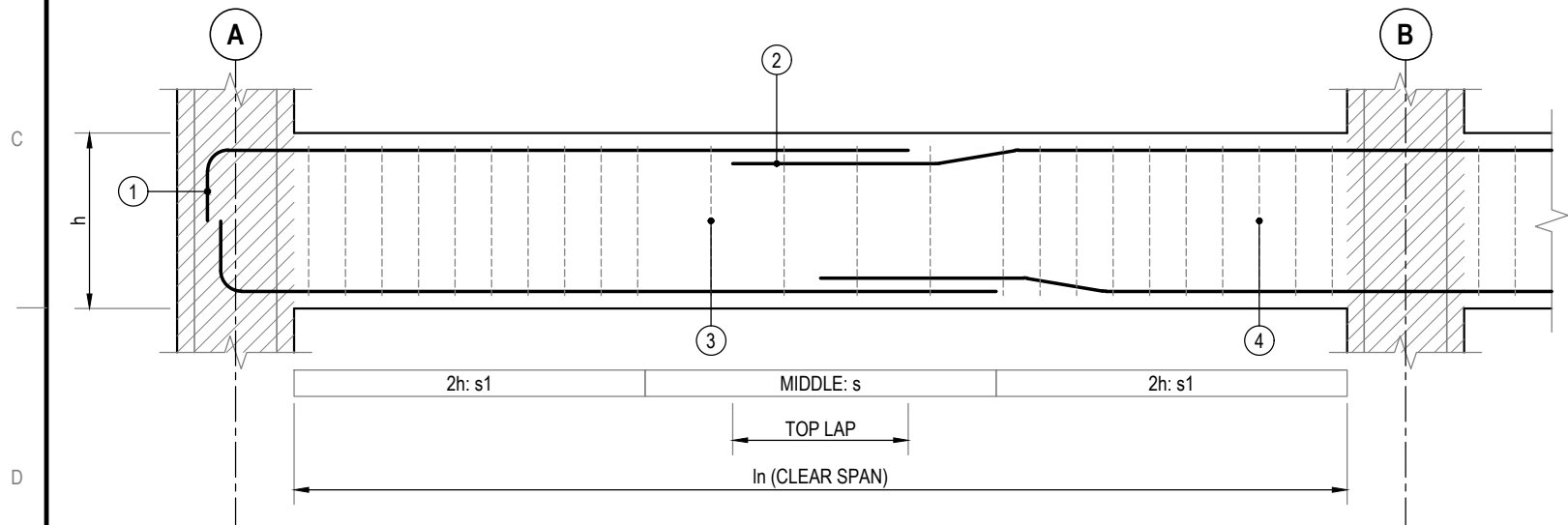




BEAM - ORDINARY MOMENT FRAME

1
1111

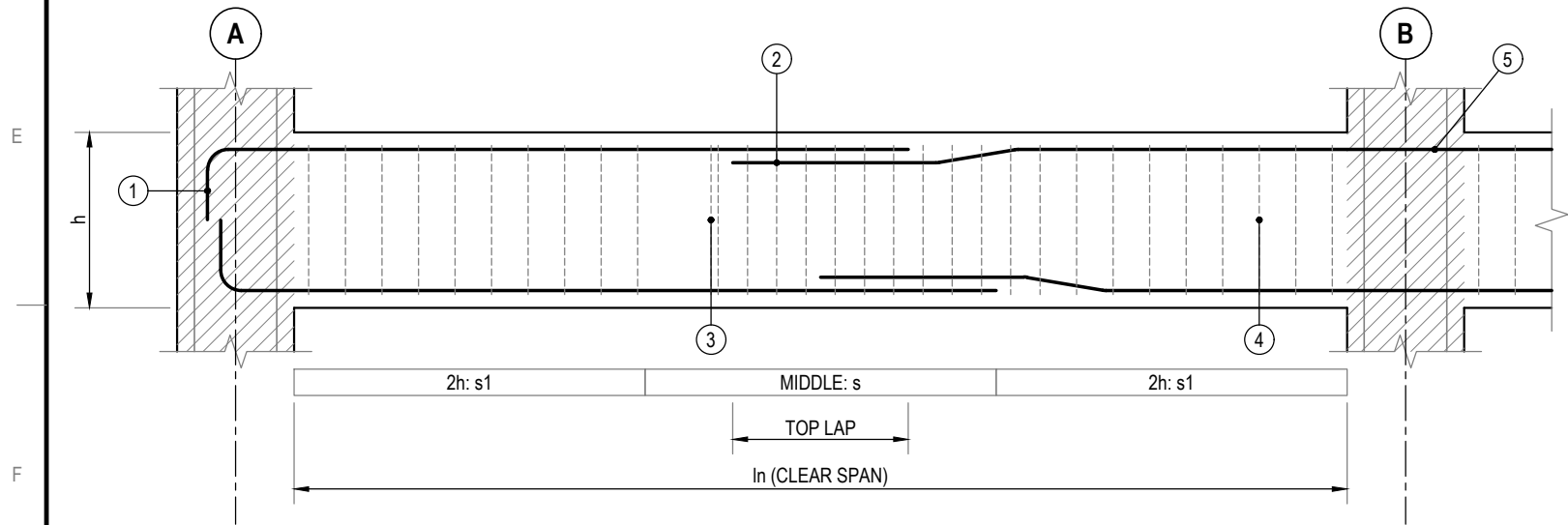
SCALE N.T.S.



BEAM - INTERMEDIATE MOMENT FRAME

2
1111

SCALE N.T.S.



BEAM - SPECIAL MOMENT FRAME

3
1111

SCALE N.T.S.

- 90° HOOKS IN THE EXTERIOR COLUMN, l_{dh} FROM THE FACE, TAILS INTO THE JOINT
- TOP BARS: ≥ 2 CONTINUOUS; CLASS B LAP NEAR MIDSPAN
- FIRST STIRRUP 50 FROM THE FACE; $s \leq d/2$, 600 ($d/4$, 300 IF $V_s > 0.33\sqrt{f_c}$ bw d)
- BOTTOM BARS: ≥ 2 CONTINUOUS; CLASS B LAP AT OR NEAR THE SUPPORT

- 90° HOOKS AT THE FAR SIDE OF THE COLUMN, l_{dh} FROM THE FACE
- TOP LAP NEAR MIDSPAN, BOTTOM LAP OUTSIDE 2h: NO LAPS WITHIN 2h OF A FACE
- MIDDLE ZONE: STIRRUPS @ $\leq d/2$
- 2h ZONES: FIRST ≤ 50 mm, $s_1 \leq d/4$, 8 db, 24 dt, 300 mm; 135° HOOKS (OR 90° + HOOK-CLIP) IN PUBLIC / DUCTILE BUILDINGS

- HOOKS INSIDE THE CONFINED CORE, $l_{dh} \geq 8$ db, 150, f_y db / $(5.3\sqrt{f_c})$
- LAPS OUTSIDE JOINTS, 2h ZONES AND HINGES; HOOPS @ $\leq d/4$, 100 mm OVER THE LAP
- MIDDLE ZONE: STIRRUPS WITH SEISMIC HOOKS BOTH ENDS @ $\leq d/2$
- 2h ZONES: CLOSED HOOPS, FIRST ≤ 50 mm, $s_1 \leq d/4$, 6 db, 24 dt, 150 mm; ALSO 2h EACH SIDE OF ANY OTHER HINGE
- BARs THROUGH THE JOINT: COLUMN ≥ 20 db OF THE LARGEST BEAM BAR

NOTES

- APPLIES TO ALL STRUCTURAL DRAWINGS.
- DIMENSIONS IN mm; LEVELS IN m.
- DO NOT SCALE FROM DRAWINGS.
- DRAFTING TO EIT 011006-19.

ISSUED FOR USE

KEY PLAN (N.T.S.)

NOT APPLICABLE
(TYPICAL DETAILS)

REVISIONS

REV	DESCRIPTION	DATE	SIGN
A	ISSUED FOR USE	03/10/2026	

OWNER / CLIENT APPROVAL

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[DESIGN OFFICE NAME]

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DRAWING TITLE

TYPICAL BEAM DETAILS (1)
STIRRUPS AND SPLICE ZONES

DESIGNED :

DRAWN :

CHECKED :

ENGINEER :

COE LICENSE NO. :

SIGNATURE :

DRAWING NO.

STR-ST-1111-F-A

SHEET 1 OF 6 | A3

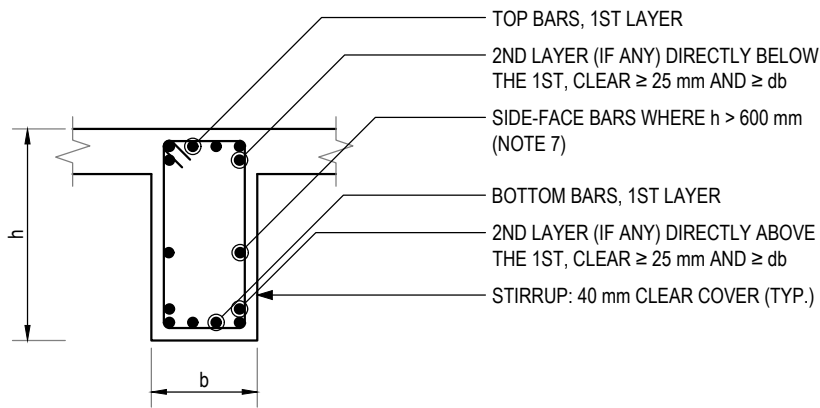
SCALE

N.T.S.

DATE

03/10/2026

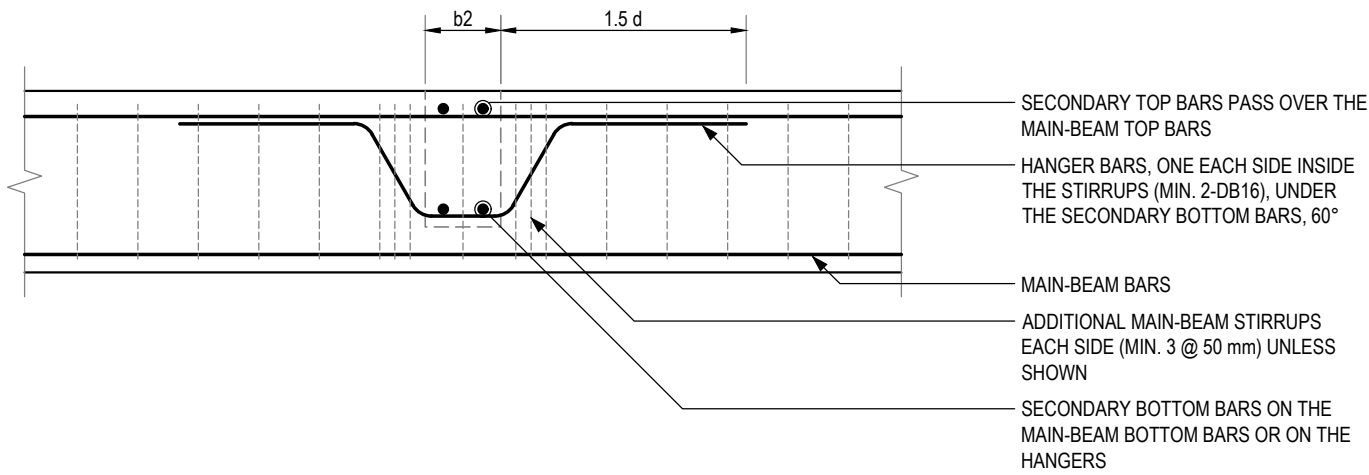
APPROVED BY : DATE :



TYPICAL BEAM SECTION

1
1113

SCALE N.T.S.

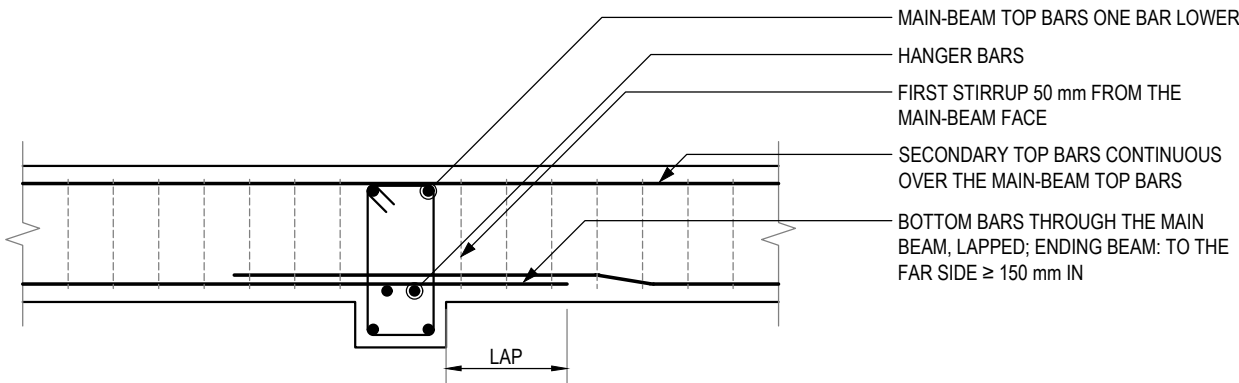


SECONDARY BEAM ON A MAIN BEAM

2
1113

SCALE N.T.S.

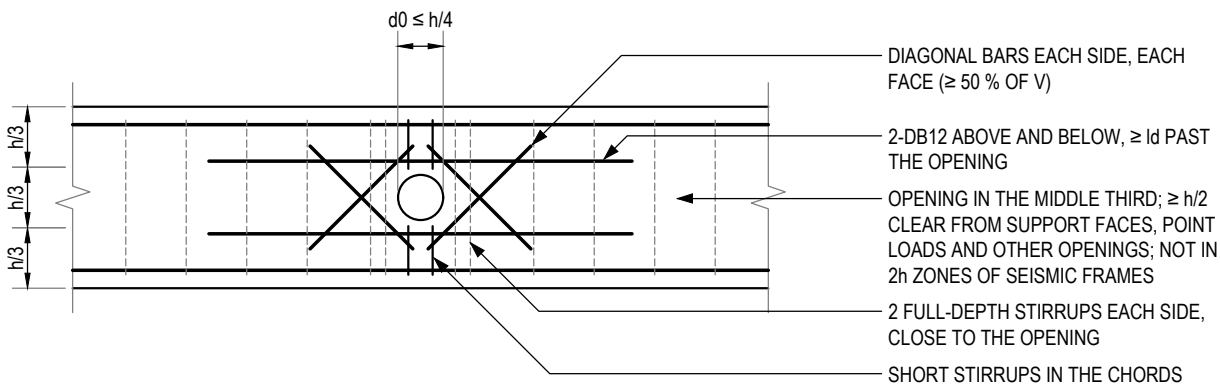
SECTION ALONG THE MAIN BEAM; SECONDARY BEAM DASHED



SECTION ALONG THE SECONDARY BEAM

A
1113

SCALE N.T.S.

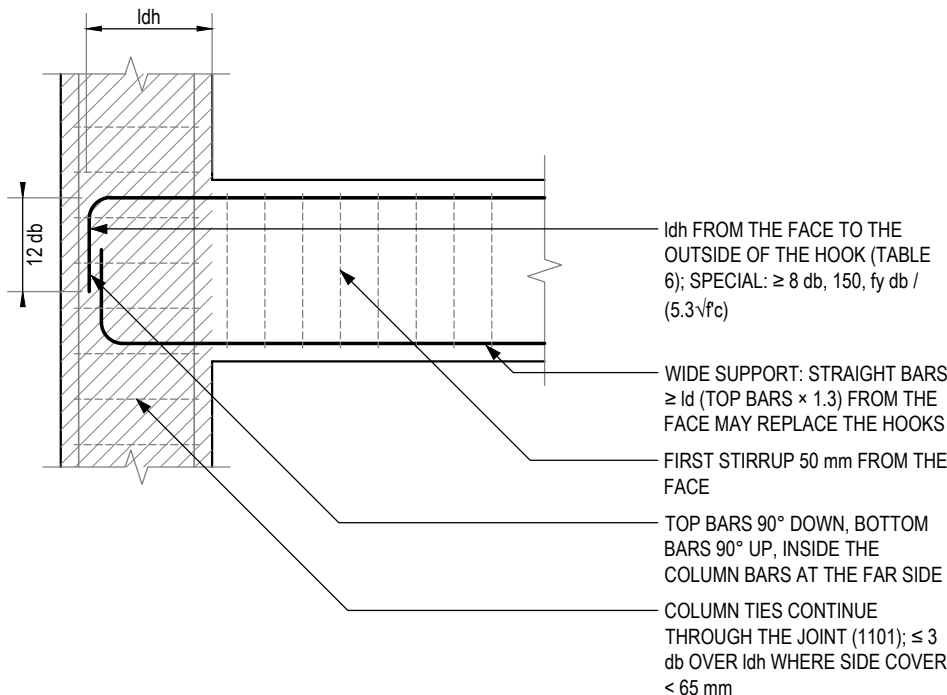


OPENING THROUGH THE WEB

3
1113

SCALE N.T.S.

LARGER OPENINGS (DEPTH ≤ h/2): PER DESIGN



BEAM END AT AN EXTERIOR COLUMN

4
1113

SCALE N.T.S.

NOTES

1. APPLIES TO ALL STRUCTURAL DRAWINGS.
2. DIMENSIONS IN mm; LEVELS IN m.
3. DO NOT SCALE FROM DRAWINGS.
4. DRAFTING TO EIT 011006-19.

ISSUED FOR USE

KEY PLAN (N.T.S.)

NOT APPLICABLE
(TYPICAL DETAILS)

REVISIONS

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DRAWING TITLE

TYPICAL BEAM DETAILS (3)
SECTIONS, SUPPORTS, OPENINGS

DESIGNED :

DRAWN :

CHECKED :

ENGINEER :

COE LICENSE NO. :

SIGNATURE :

DRAWING NO.

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SHEET 3 OF 6 | A3

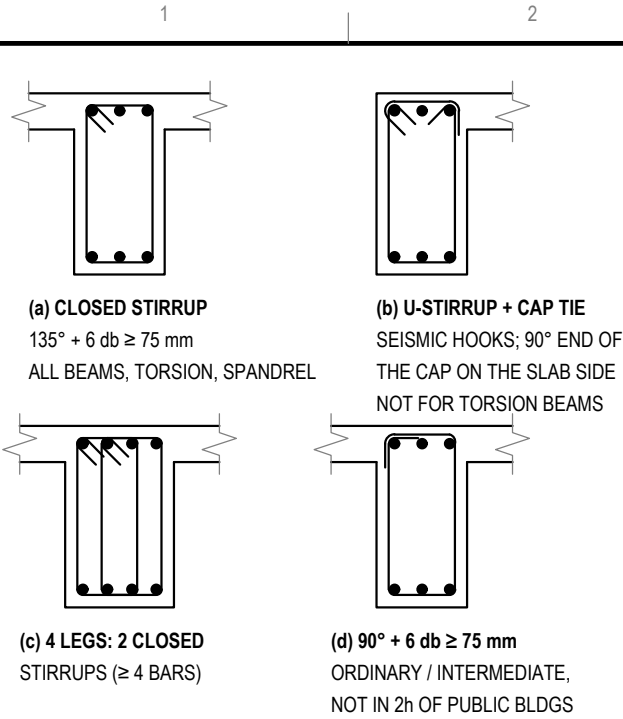
SCALE

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DATE

03/10/2026

APPROVED BY : DATE :



STIRRUP AND HOOP TYPES

SCALE N.T.S.

1116

TABLE 15 - BEAM REINFORCEMENT BY FRAME TYPE

ITEM	ORDINARY	INTERMEDIATE	SPECIAL
USE (DPT T2.3-1)	SEISMIC CATEGORY B ONLY (R = 3)	CATEGORY B, C; D ONLY ≤ 40 m, FORCES + 40 % (R = 5)	ALL CATEGORIES (R = 8)
SECTION	$h \geq \ell/16, \ell/18.5, \ell/21, \ell/8$ (NO DEFLECTION CHECK)	AS ORDINARY	$\ell_n \geq 4 d$; $b_w \geq$ LESSER OF 0.3 h AND 250 mm; WIDTH ≤ COLUMN + LESSER OF c_2 , $\frac{3}{4} c_1$ EACH SIDE
LONGITUDINAL BARS	≥ 2 TOP + 2 BOTTOM CONTINUOUS; $A_s \geq 1.4 b_w d / f_y$	+Mn AT FACE ≥ 1/3 -Mn; ANY SECTION ≥ 1/5 MAX	≥ 2 TOP + 2 BOTTOM CONTINUOUS; $\rho \leq 0.025$; +Mn AT FACE ≥ ½ -Mn; ANY SECTION ≥ ¼ MAX
END ZONE	-	2h FROM EACH FACE	2h FROM EACH FACE AND EACH SIDE OF ANY HINGE
FIRST STIRRUP	50 mm FROM THE FACE	≤ 50 mm FROM THE FACE	≤ 50 mm FROM THE FACE
SPACING IN 2h	-	$s_1 \leq \min (d/4, 8 \text{ db}, 24 \text{ dt}, 300)$	CLOSED HOOPS, $s_1 \leq \min (d/4, 6 \text{ db}, 24 \text{ dt}, 150)$; STRICTER OF DPT AND ACI 318
SPACING ELSEWHERE	$s \leq d/2, 600 (d/4, 300 \text{ IF } V_s > 0.33\sqrt{f_c} b_w d)$	≤ d/2	≤ d/2, SEISMIC HOOKS BOTH ENDS
HOOKS	90° + 6 db (≤ DB16); 135° + 6 db FOR CLOSED / TORSION STIRRUPS	90° + 6 db ≥ 75 mm; 135° OR HOOK-CLIP IN 2h FOR PUBLIC / DUCTILE BUILDINGS (DPT; ACI 318: HOOPS)	135° + 6 db ≥ 75 mm; CAP TIE 135° / 90°, 90° ENDS ALTERNATE (ALL ON THE SLAB SIDE IF ONE SLAB)
LAP LOCATION	TOP NEAR MIDSPAN; BOTTOM AT OR NEAR THE SUPPORT	AS ORDINARY, BUT NOT WITHIN 2h	NOT IN JOINTS, 2h ZONES OR HINGES; HOOPS @ ≤ d/4, 100 mm OVER THE LAP
EXTERIOR JOINT	BARs ANCHORED FOR f_y : 90° HOOK, ℓ_{dh} FROM THE FACE	90° HOOKS AT THE FAR SIDE, ℓ_{dh} FROM THE FACE	TO THE FAR FACE OF THE CORE; $\ell_{dh} \geq 8 \text{ db}, 150, f_y \text{ db} / (5.3\sqrt{f_c})$
REFERENCE	EIT 011008 7.13, 9.5, 11.4, 12.9 – 12.14; DPT 5.2.6	DPT 1301/1302 5.2.7.3, 5.2.7.6	DPT 1301/1302 5.2.8, 5.2.10

TYPICAL BEAM NOTES

- THESE DETAILS APPLY WHERE THE BEAM DRAWINGS DO NOT SHOW OTHERWISE. FRAME TYPE AND SEISMIC CATEGORY: SEE THE DESIGN CRITERIA; REINFORCEMENT BY FRAME TYPE: TABLE 15.
- COVER TO STIRRUPS: 40 mm INTERIOR; 50 mm (≥ DB20) / 40 mm (≤ DB16) EXPOSED; 75 mm CAST AGAINST EARTH (TABLE 7).
- AT LEAST ONE BAR IN EACH CORNER; ≥ 2 TOP AND 2 BOTTOM BARS CONTINUOUS OVER THE FULL LENGTH, ANCHORED AT END SUPPORTS. PERIMETER BEAMS: CONTINUOUS TOP ≥ 1/6 OF THE SUPPORT TOP BARS (LAP AT MIDSPAN), CONTINUOUS BOTTOM ≥ 1/4 OF THE MIDSPAN BOTTOM BARS (LAP AT THE SUPPORT), CLASS B, CLOSED STIRRUPS OVER THE WHOLE SPAN; INTEGRITY BARS PASS INSIDE THE COLUMN BARS. [EIT 011008 7.13.2; DPT 5.2.6]
- CLEAR SPACING ≥ db, 25 mm AND 4/3 MAX. AGGREGATE. LAYERS DIRECTLY ABOVE EACH OTHER, CLEAR ≥ 25 mm AND ≥ db. [EIT 011008 7.6]
- STIRRUPS CLOSED, HOOKS AT THE TOP (SLAB SIDE); FIRST STIRRUP 50 mm FROM THE SUPPORT FACE. STIRRUPS ENCLOSE COMPRESSION BARS AT $s \leq 16 \text{ db}, 48 \text{ dt}$. SPECIAL-FRAME HOOP ZONES: EVERY CORNER AND ALTERNATE BAR HELD BY A HOOP CORNER OR CROSSTIE, NO BAR > 150 mm CLEAR FROM A HELD BAR. INTERMEDIATE AND SPECIAL FRAMES: DEFORMED HOOPS AND CROSSTIES, DB10 MIN. SPECIAL-FRAME HOOPS: THE STRICTER OF DPT 5.2.8.3.2 (d/4, 8 db, 24 dt, 300) AND ACI 318-11 21.5.3.2 (d/4, 6 db, 150). [EIT 011008 7.11, 7.10.5.3; DPT 5.2.8.3.3]
- LAPS: CLASS B PER TABLE 6, ADJACENT LAPS STAGGERED ≥ 1.0 m; NO LAPS FOR BARS > DB36. COUPLERS ≥ 1.25 f_y . [EIT 011008 12.13, 12.14]
- BAR ENDS ON THE ELEVATIONS: A PLAIN END (NO MARK) MEANS THE BAR STOPS THERE (END OF AN ADDITIONAL OR LAPPED BAR); NO MARK AT A BREAK LINE MEANS THE BAR CONTINUES. KEY ON 1112.
- CUT-OFF AND CANTILEVER: 1112. SECTIONS, EDGE AND TORSION BEAMS, SIDE-FACE BARS, SECONDARY BEAMS, END ANCHORAGE AND OPENINGS: 1113. LEVEL CHANGES AND STEPS: 1114. SCHEDULE: 1115. NO CORING OR SLEEVES THROUGH BEAMS WITHOUT THE ENGINEER'S APPROVAL.

- NOTES
- APPLIES TO ALL STRUCTURAL DRAWINGS.
 - DIMENSIONS IN mm; LEVELS IN m.
 - DO NOT SCALE FROM DRAWINGS.
 - DRAFTING TO EIT 011006-19.

ISSUED FOR USE

KEY PLAN (N.T.S.)

NOT APPLICABLE
(TYPICAL DETAILS)

REVISIONS

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DRAWING TITLE

TYPICAL BEAM DETAILS (6)
STIRRUP TYPES, TABLE, NOTES

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DRAWN :
CHECKED :
ENGINEER :
COE LICENSE NO. :
SIGNATURE :

DRAWING NO.
STR-ST-1116-F-A
SHEET 6 OF 6 | A3

SCALE
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DATE
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